

Demonstrations

- Mt. Mount Athos (a location based use case)
- Mt. Voras and Mt. Paiko (a taxon/location based exploration)
- Norway's Ulva (a taxon/location based exploration)
- Cretan Sea (a location based use case)

The Mt. Athos use case

<!-- Demo as back-up slides starts here -->

Exploring Molecular Biodiversity

Input data sources

☒ ENA

☐ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01

to

2025-06-21

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

Load Data

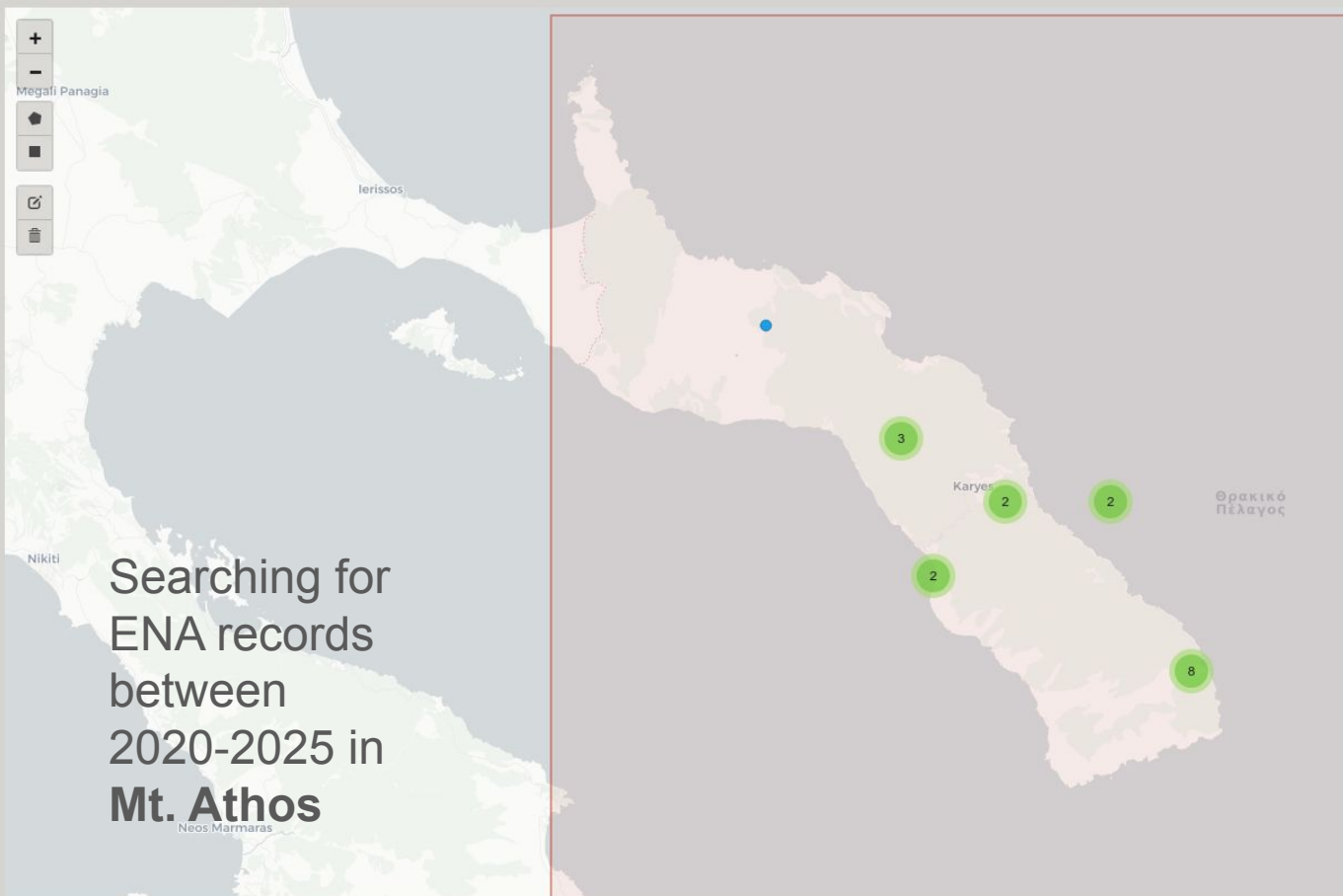
Table options

☐ Show filter

Group by

☐ Tax_division

☐ Scientific_name





Input data sources

☒ ENA☐ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01

to

2025-06-21

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

Load Data

Table options

☐ Show filter

Group by

☐ Tax_division☐ Scientific_name☐ Tag1

Home Overview Table Map

ENA

accession	first_public	country	altitude	host	host_tax_id	isolation_source	scientific_name	tax_id	lat	long	tax_division2	tag1	tag2
QQ519878	2023-09-13	Greece					Euphorbia oblongata	212945	40.33333333	24.11666667	Plantae		
ON259090	2022-11-29	Greece: Athos Peninsula, Panteleimons Monastery					Lathyrus oleraceus subsp. biflorus	47742	40.28	24.2	Plantae		
ON259089	2022-11-29	Greece: Athos Peninsula, 1 km S of Daphne					Lathyrus oleraceus subsp. biflorus	47742	40.2	24.22	Plantae		
ON243976	2022-11-27	Greece: Athos Peninsula, below Karyes					Lathyrus oleraceus subsp. biflorus	47742	40.26	24.25	Plantae		
ON243975	2022-11-27	Greece: Athos Peninsula, Xeropotamou Monastery					Lathyrus oleraceus subsp. biflorus	47742	40.23	24.22	Plantae		
ON243977	2022-11-27	Greece: Athos Peninsula,					Lathyrus oleraceus subsp.	47742	40.24	24.28	Plantae		

Found 9
records of
*Lathyrus
oleraceus*
subsp. *biflorus*

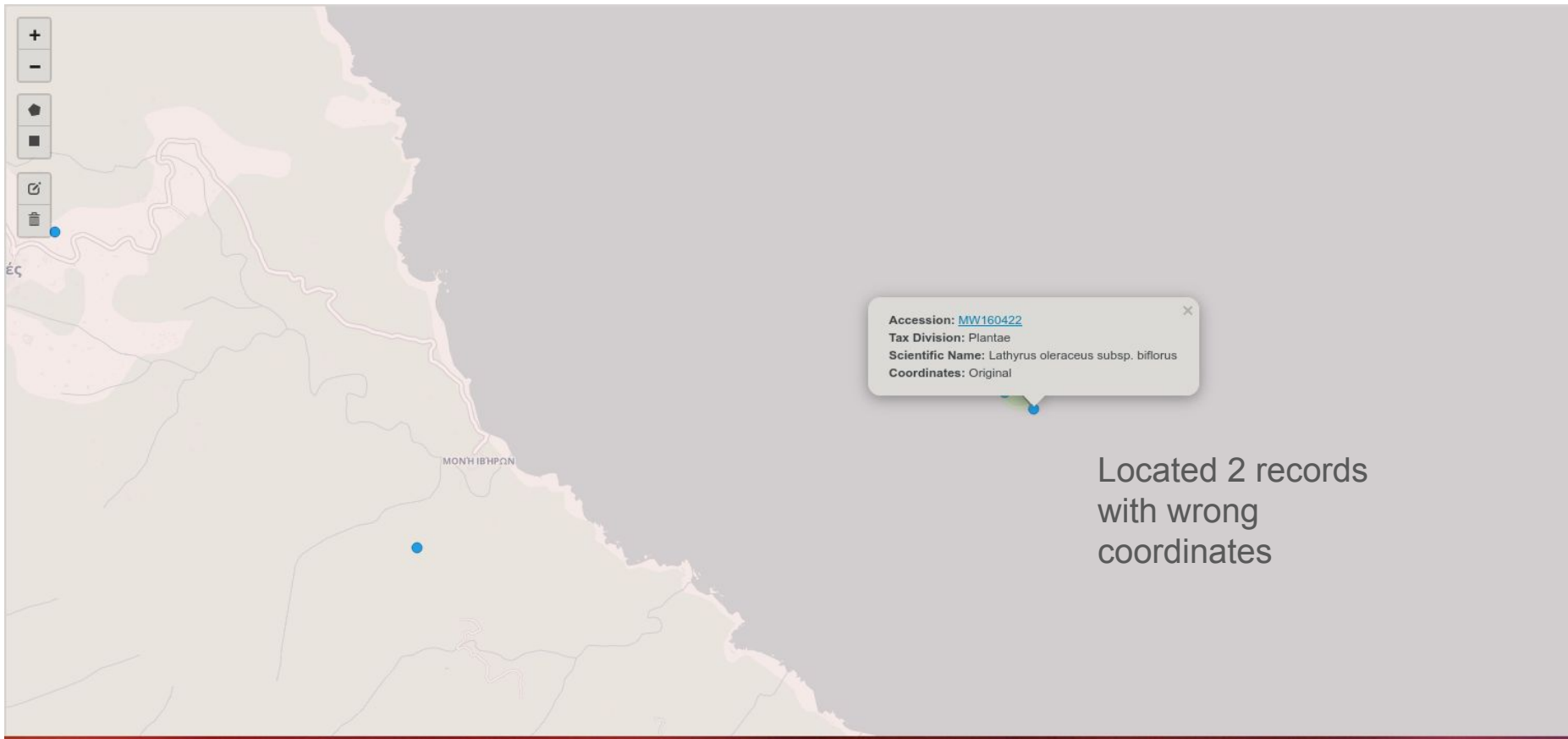
⚙ Most Visited [f Fedora Docs](#) [FM Fedora Magazine](#) [Fedora Project](#) [User Commu](#)

```
ID  ON259090; SV 1; circular; genomic DNA; STD; PLN; 121410 BP.
XX
AC  ON259090;
XX
DT  29-NOV-2022 (Rel. 144, Created)
DT  30-NOV-2022 (Rel. 144, Last updated, Version 1)
XX
DE  Pisum sativum subsp. elatius voucher PI344009 chloroplast, complete genome.
XX
KW  .
XX
OS  Pisum sativum subsp. elatius
OC  Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;
OC  Spermatophyta; Magnoliopsida; eudicotyledons; Gunneridae; Pentapetalae;
OC  rosids; fabids; Fabales; Fabaceae; Papilionoideae; 50 kb inversion clade;
OC  NPAAA clade; Hologalegina; IRL clade; Fabeae; Pisum.
OG  Plastid:chloroplast
XX
```

Each record corresponds to a full plastome,
very useful for studying phylogenetic scenarios
and gene flow via seed.



Lathyrus oleraceus



Located 2 records
with wrong
coordinates

Exploring Molecular Biodiversity

Home Overview Table Map

Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01 to 2025-06-21

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

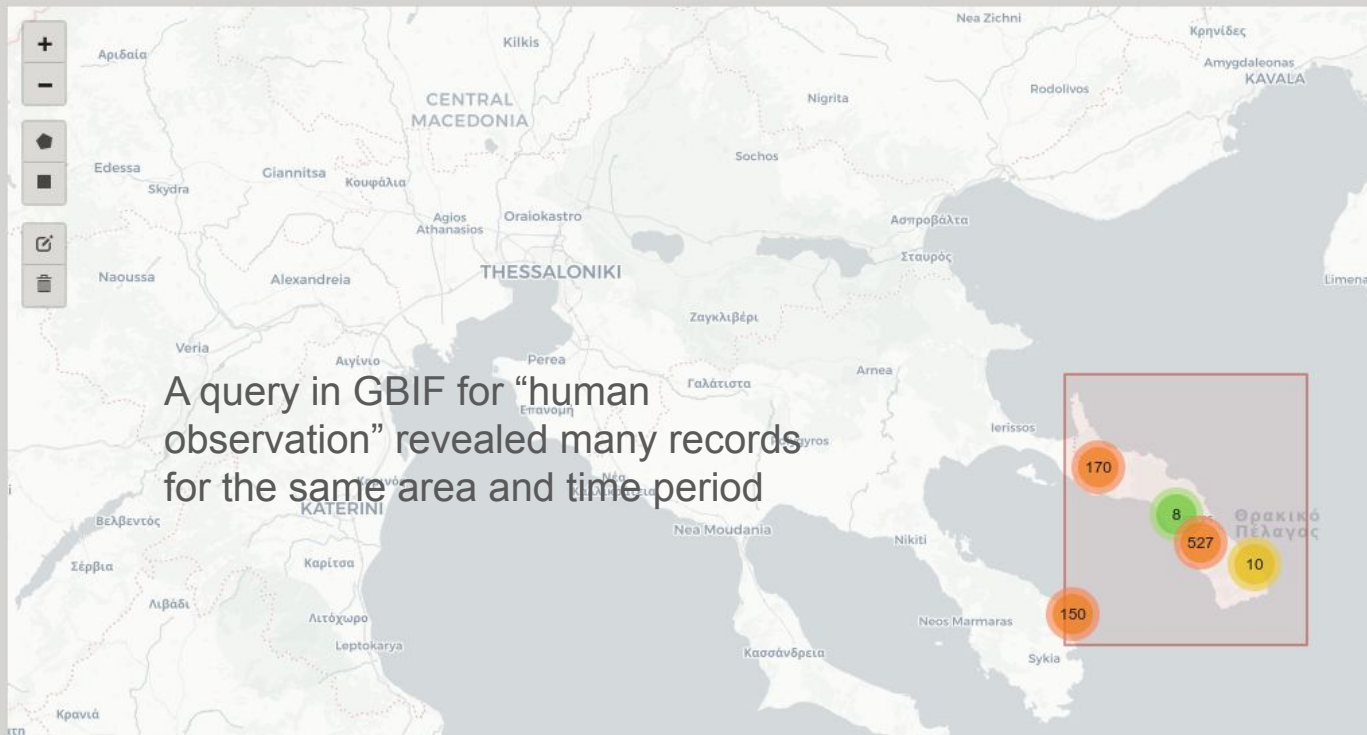
GBIF basisOfRecord

MATERIAL_SAMPLE
OBSERVATION
HUMAN_OBSERVATION

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 23.983, E 24.478, S 40.059, N 40.48 All points





Country of interest: <

Greece ▼

Dates of interest:

2020-05-01 to 2025-06-21

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE
OBSERVATION
HUMAN_OBSERVATION

Load Data

Table options

☒ Show filter

Group by

☒ Tax_division
☐ Scientific_name
☐ Tag1
☐ Tag2

[Home](#) [Overview](#) [Table](#) [Map](#)

ENA

GBIF

tax_division2	accession	scientific_name	country	first_public	basis_of_record	decimalLatitude	decimalLongitude	phylum	class	order	family	genus	species
▶ Animalia (522)													
▶ Plantae (323)													
▶ Chromista (1)													
▶ Fungi (1)													
1–4 of 4 rows Show 15 ▼													

Previous 1 of 1 Next

[Download as CSV](#)



Country of interest: <

Greece

Dates of interest:

2020-05-01 to 2025-06-21

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE
OBSERVATION
HUMAN_OBSERVATION

Load Data

Table options

☒ Show filter

Group by

- ☐ Tax_division
- ☐ Scientific_name
- ☐ Tag1
- ☐ Tag2

ENA		GBIF											
accession	scientific_name	country	first_public	basis_of_record	decimalLatitude	decimalLongitude	tax_division2	phylum	class	order	family	genus	species
									Inse				
5205976447	Nemoptera sinuata	Greece	2025-06-19	HUMAN_OBSERVATION	40.226456	24.238582	Animalia	Arthropoda	Insecta	Neuroptera	Nemopteridae	Nemoptera	Nemoptera sinuata
6185420478	Nordmannia ilicis	Greece	2025-06-18	HUMAN_OBSERVATION	40.251298	24.261515	Animalia	Arthropoda	Insecta	Lepidoptera	Lycaenidae	Nordmannia	Nordmannia ilicis
5288091168	Tettigonia viridissima	Greece	2025-06-16	HUMAN_OBSERVATION	40.171564	24.382187	Animalia	Arthropoda	Insecta	Orthoptera	Tettigoniidae	Tettigonia	Tettigonia viridissima
5867986760	Paracorymbia pallens	Greece	2025-05-27	HUMAN_OBSERVATION	40.251863	24.252543	Animalia	Arthropoda	Insecta	Coleoptera	Cerambycidae	Paracorymbia	Paracorymbia pallens
5868018099	Lymantria dispar	Greece	2025-05-27	HUMAN_OBSERVATION	40.215075	24.22257	Animalia	Arthropoda	Insecta	Lepidoptera	Erebidae	Lymantria	Lymantria dispar
5868194443	Cataglyphis nodus	Greece	2025-05-27	HUMAN_OBSERVATION	40.250625	24.258252	Animalia	Arthropoda	Insecta	Hymenoptera	Formicidae	Cataglyphis	Cataglyphis nodus
5868197685	Regiscolia Betrem & Bradley, 1964	Greece	2025-05-27	HUMAN_OBSERVATION	40.197614	24.324811	Animalia	Arthropoda	Insecta	Hymenoptera	Scoliidae	Regiscolia	
5868411682	Lymantria dispar	Greece	2025-05-27	HUMAN_OBSERVATION	40.215075	24.22257	Animalia	Arthropoda	Insecta	Lepidoptera	Erebidae	Lymantria	Lymantria dispar
5935655736	Melitaea didyma	Greece	2025-05-27	HUMAN_OBSERVATION	40.215237	24.222897	Animalia	Arthropoda	Insecta	Lepidoptera	Nymphalidae	Melitaea	Melitaea didyma
6131561725	Melitaea didyma	Greece	2025-05-27	HUMAN_OBSERVATION	40.215237	24.222897	Animalia	Arthropoda	Insecta	Lepidoptera	Nymphalidae	Melitaea	Melitaea didyma
4854037805	Stenbanus	Greece	2024-08-16	HUMAN_OBSERVATION	40.267975	24.27715	Animalia	Arthropoda	Insecta	Hymenoptera	Stenbanidae	Stenbanus	Stenbanus

Filter for insects

What we can do here:

- Compare full plastomes for all *Lathyrus oleraceus* records and find spatial genetic patterns
 - Come up with possible evolutionary forces defining these patterns
 - Come up with relevant conservation strategies
- Evaluate specific habitats with the help of both ENA and GBIF records
 - Describe the environment of the habitat for the plant we focus.
 - Align spatial patterns between genetic diversity of *Lathyrus* and species distribution of birds, insects.

The Mt. Athos use case

<!-- Demo as back-up slides ends here -->

The Mt. Voras and Mt. Paiko use case

<!-- Demo as back-up slides starts here -->

Exploring Molecular Biodiversity

Input data sources

☒ ENA

☐ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01

to

2025-06-21

Scientific name (optional)

Fagus

Kingdom (optional)

Load Data

Home Overview Table Map

Welcome

Welcome to Odyssey, an interactive

Who is the app intended for?

The app provides a user-friendly interface

Methodology

The current app prototype queries ENA and GBIF

It provides tools for data exploration and visualization

The modular design ensures flexibility and scalability

Contribution

Your input is invaluable - whether it's a new record or a correction

Feel free to open a [GitHub issue](#) or contact the developers

License

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The code contained in this app is licensed under the MIT License

Searching for
ENA records
between
2020-2025
about the
beech tree,
Fagus sp.



Fagus sylvatica

ENA

accession	first_public	country	altitude	host	host_tax_id	isolation_source	scientific_name	tax_id	lat	long	tax_division2
OR936132	2023-12-25	Greece: Voras, NW					Fagus sylvatica	28930	39.0742	21.8243	Plantae
OR936129	2023-12-25	Greece: Paiko, NW					Fagus sylvatica	28930	39.0742	21.8243	Plantae

 Download as CSV

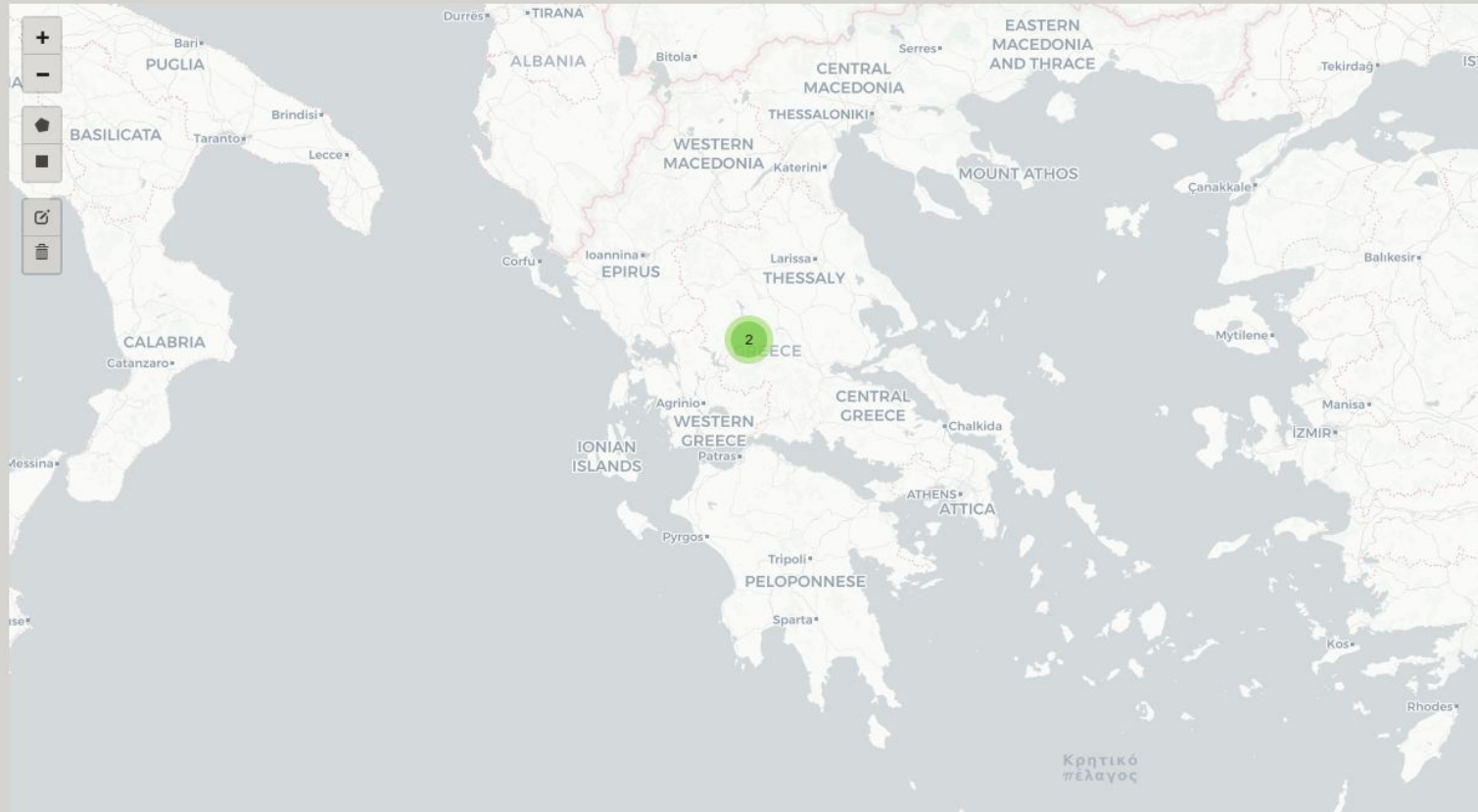
```
ID OR936132; SV 1; circular; genomic DNA; STD; PLN; 158449 BP.
XX
AC OR936132;
XX
DT 25-DEC-2023 (Rel. 144, Created)
DT 26-DEC-2023 (Rel. 144, Last updated, Version 1)
XX
DE Fagus sylvatica chloroplast, complete genome.
XX
KW .
XX
OS Fagus sylvatica (European beech)
OC Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;
OC Spermatophyta; Magnoliopsida; eudicotyledons; Gunneridae; Pentapetalae;
OC rosids; fabids; Fagales; Fagaceae; Fagus.
OG Plastid:chloroplast
XX
```

Only two records, which are full plastomes, without any geographical coordinates

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

No area filter selected 2 points have estimated coordinates (2 Greece)



Since there are no coordinates, these records are placed in the center of Greece

ENA

accession	first_public	country	altitude	host	host_tax_id	isolation_source	scientific_name	tax_id	lat	long	tax_division2
OR936132	2023-12-25	Greece: Voras, NW					Fagus sylvatica	28930	39.0742	21.8243	Plantae
OR936129	2023-12-25	Greece: Paiko, NW					Fagus sylvatica	28930	39.0742	21.8243	Plantae

[Download as CSV](#)

```
ID  OR936132; SV 1; circular; genomic DNA; STD; PLN; 158449 BP.
XX
AC  OR936132;
XX
DT  25-DEC-2023 (Rel. 144, Created)
DT  26-DEC-2023 (Rel. 144, Last updated, Version 1)
XX
DE  Fagus sylvatica chloroplast, complete genome.
XX
KW  .
XX
OS  Fagus sylvatica (European beech)
OC  Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;
OC  Spermatophyta; Magnoliopsida; eudicotyledons; Gunneridae; Pentapetales;
OC  rosids; fabids; Fagales; Fagaceae; Fagus.
OG  Plastid:chloroplast
XX
```

However, we can find the locations of these samples (under “country”) on Mt. Paiko and Mt. Voras, two mountains that form one of the most important glacial refugia in Europe

Exploring Molecular Biodiversity

Home Overview Table Map

Input data sources

☒ ENA

☐ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01 to 2025-06-21

Scientific name (optional)

Kingdom (optional)

Plantae

Load Data

Table options

☐ Show filter

Group by

☐ Tax_division

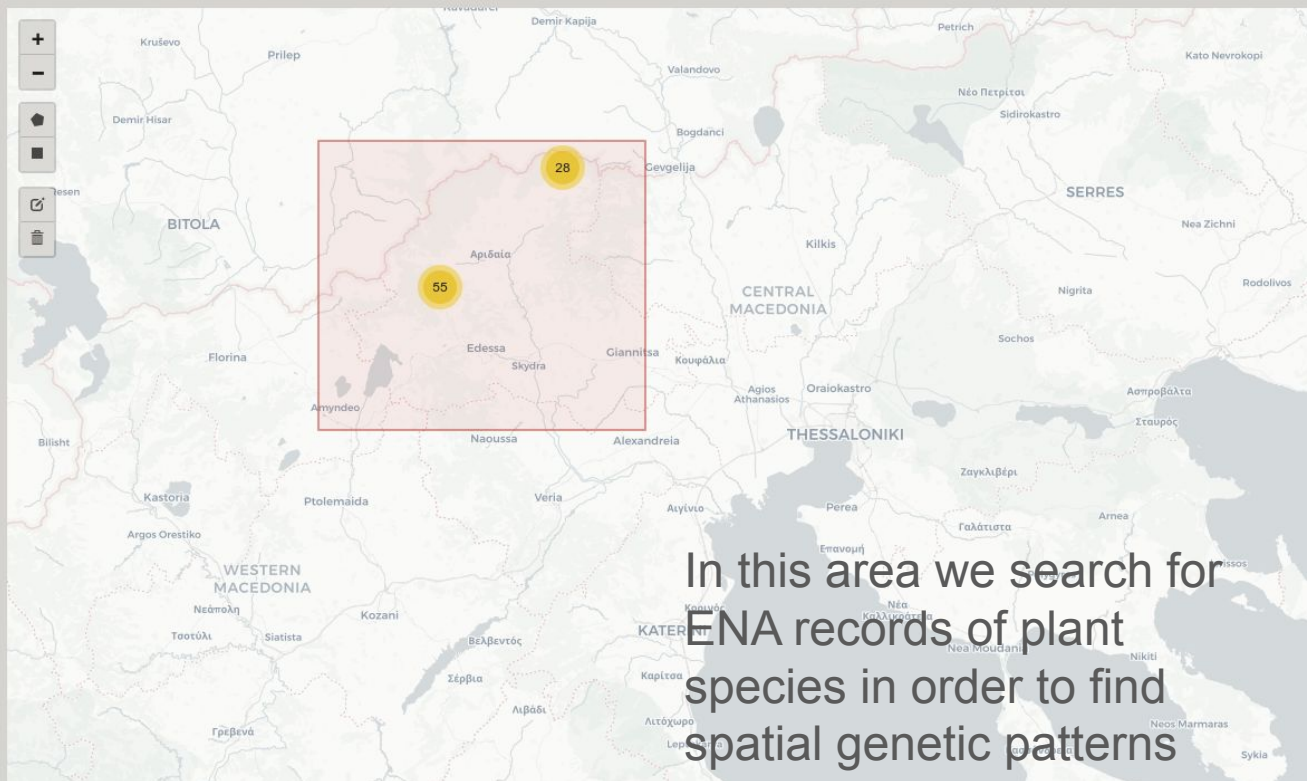
☐ Scientific_name

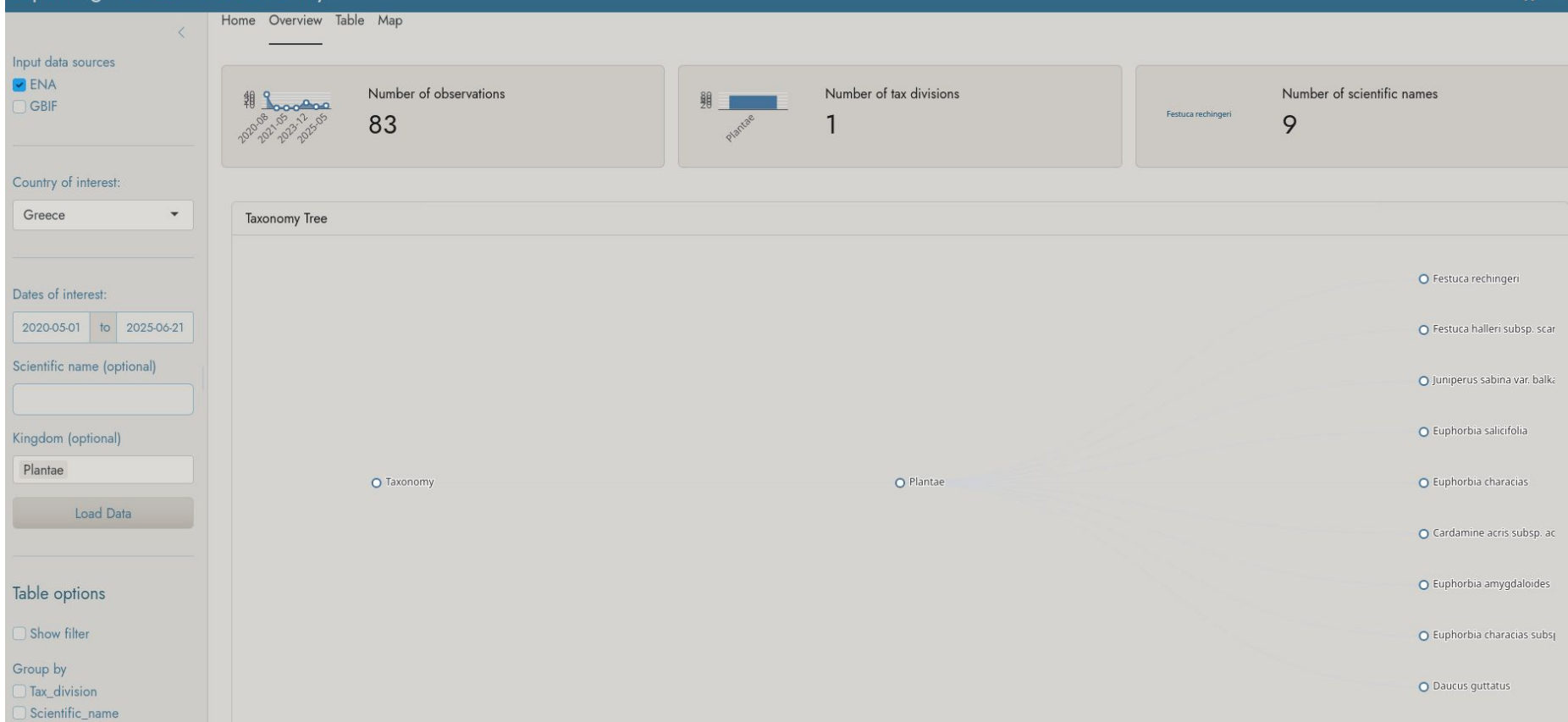
☐ Tax1

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 21.638, E 22.442, S 40.648, N 41.185 All points use original coordinate





We can select one of the plant species found on the database to further investigate their spatial genetic structure in this area.

ID PP337632; SV 1; linear; genomic DNA; STD; PLN; 726 BP.
XX
AC PP337632;
XX
DT 09-MAY-2025 (Rel. 144, Created)
DT 31-AUG-2025 (Rel. 144, Last updated, Version 4)
XX
DE Festuca halleri subsp. scardica isolate Fb579 trnA-Leu (trnL) gene and
DE trnL-trnF intergenic spacer, partial sequence; chloroplast.
XX



OQ818780; SV 1; linear; genomic DNA; STD; PLN; 644 BP.

OQ818780;

20-DEC-2023 (Rel. 144, Created)
21-DEC-2023 (Rel. 144, Last updated, Version 1)

Juniperus sabina var. balkanensis isolate GR1-3 trnD-trnT intergenic spacer
and tRNA-Thr (trnT) gene, partial sequence; chloroplast.



ID OQ230305; SV 1; linear; genomic DNA; STD; PLN; 1511 BP.
XX
AC OQ230305;
XX
DT 14-JUN-2023 (Rel. 144, Created)
DT 15-JUN-2023 (Rel. 144, Last updated, Version 1)
XX
DE Euphorbia characias isolate 14772 trnT-trnL intergenic spacer, partial
DE sequence; tRNA-Leu (trnL) gene, complete sequence; and trnL-trnF intergenic
DE spacer, partial sequence; plastid.
XX

Exploring Molecular Biodiversity

Home Overview Table Map

Input data sources

☐ ENA

☒ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01 to 2025-06-21

Scientific name (optional)

Kingdom (optional)

GBIF basisOfRecord

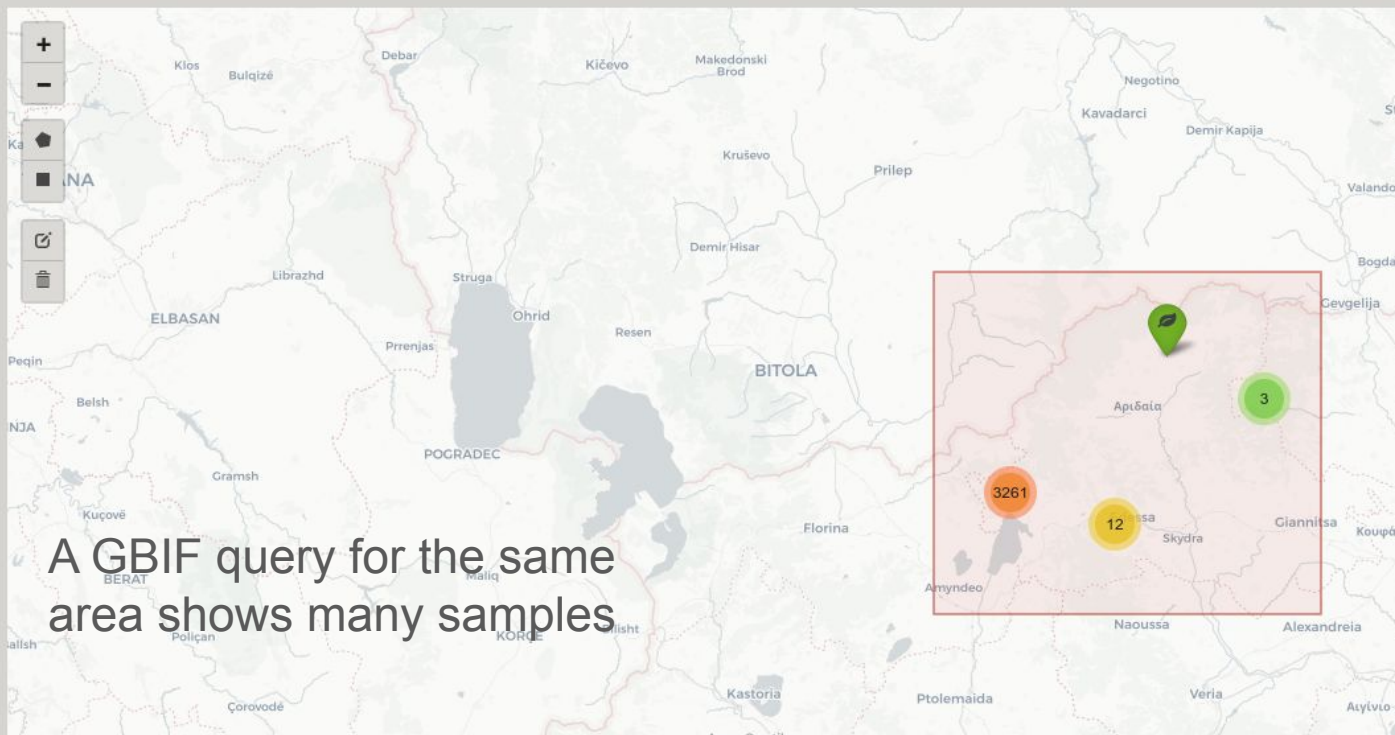
MATERIAL_SAMPLE

Load Data

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 21.638, E 22.442, S 40.648, N 41.185 All points use o



Exploring Molecular Biodiversity

Input data sources

☐ ENA

☒ GBIF

Country of interest:

Greece

Dates of interest:

2020-05-01

to

2025-06-21

Scientific name (optional)

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE

Load Data

Table options

☐ Show filter

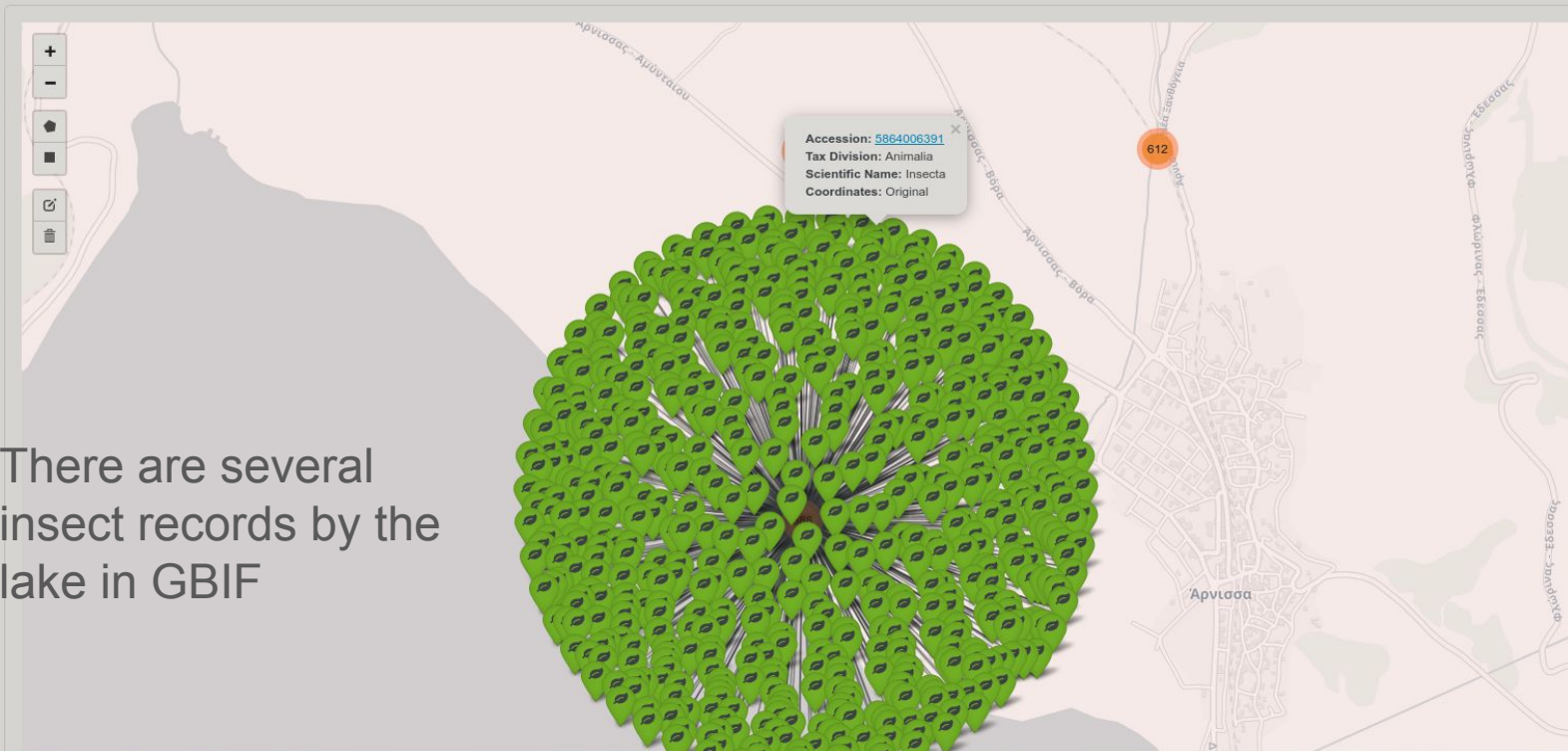
Group by

Home Overview Table Map

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 21.638, E 22.442, S 40.648, N 41.185 All points use original coordinates



What we can do here:

- Compare different plants on their genetic diversity in specific barcoding regions.
 - Identify spatial patterns and possible evolutionary forces defining these patterns
 - Come up with relevant conservation strategies
- Evaluate specific habitats with the help of GBIF insect records
 - Describe the insect fauna of a glacial refugium and the spatial patterns within the area
 - Evaluate environmental conditions in the habitats and identify threats
 - Conclude with conservation strategies to address these threats

The Mt. Voras and Mt. Paiko use case

<!-- Demo as back-up slides ends here -->

The Norway *Ulva* use case

<!-- Demo as back-up slides starts here -->

Exploring Molecular Biodiversity



Home Overview Table Map

Input data sources

☒ ENA

☐ GBIF

Country of interest:

Norway

Dates of interest:

2023-05-04

to

2025-06-20

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

Plantae

Load Data

Table options

☐ Show filter

Group by

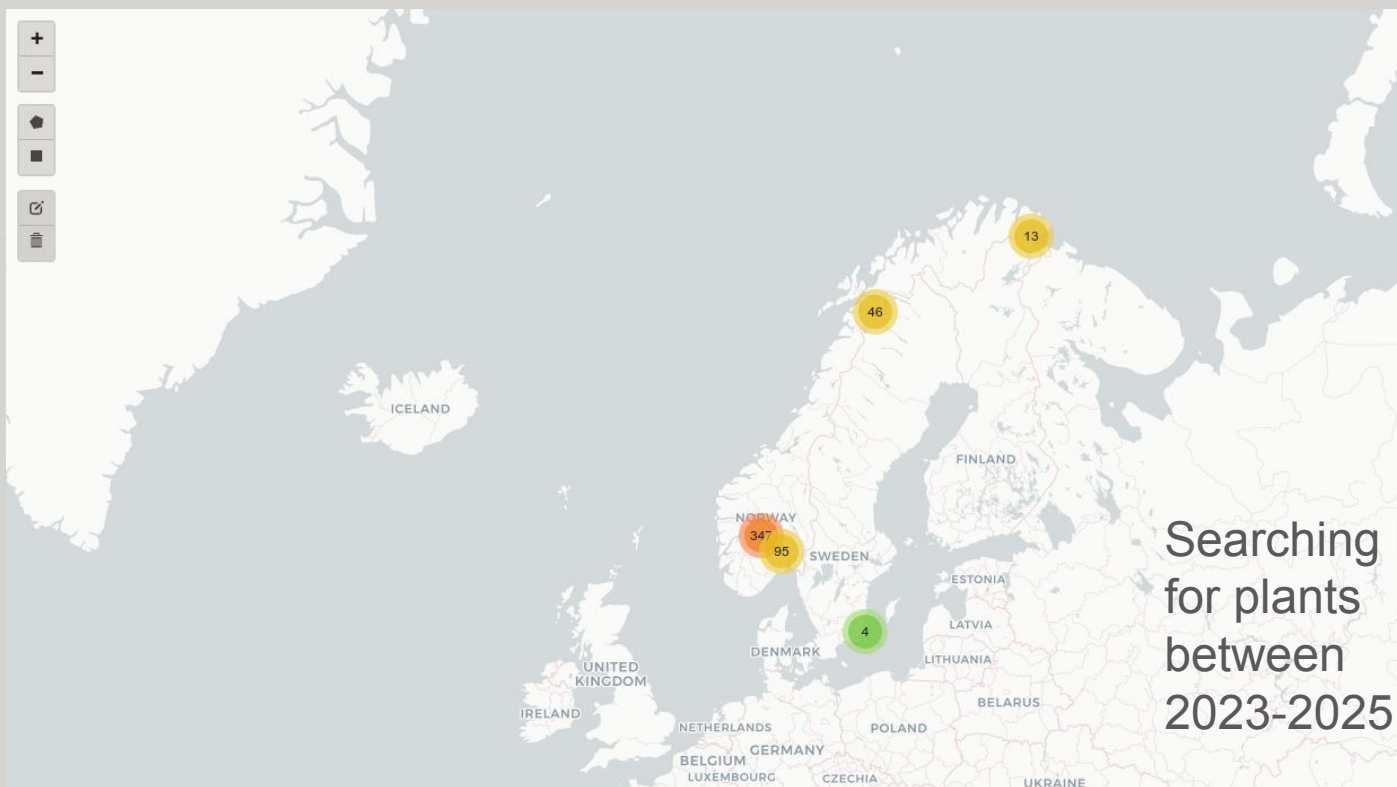
☐ Tax_division

☐ Scientific_name

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

No area filter selected 399 points have estimated coordinates (347 Norway, 52 Greece)



Input data sources

☒ ENA☐ GBIF

Country of interest:

Norway

Dates of interest:

2023-05-04

to

2025-06-20

Scientific name (optional)

Ulva

Kingdom (optional)

Load Data

Table options

☐ Show filter

Group by

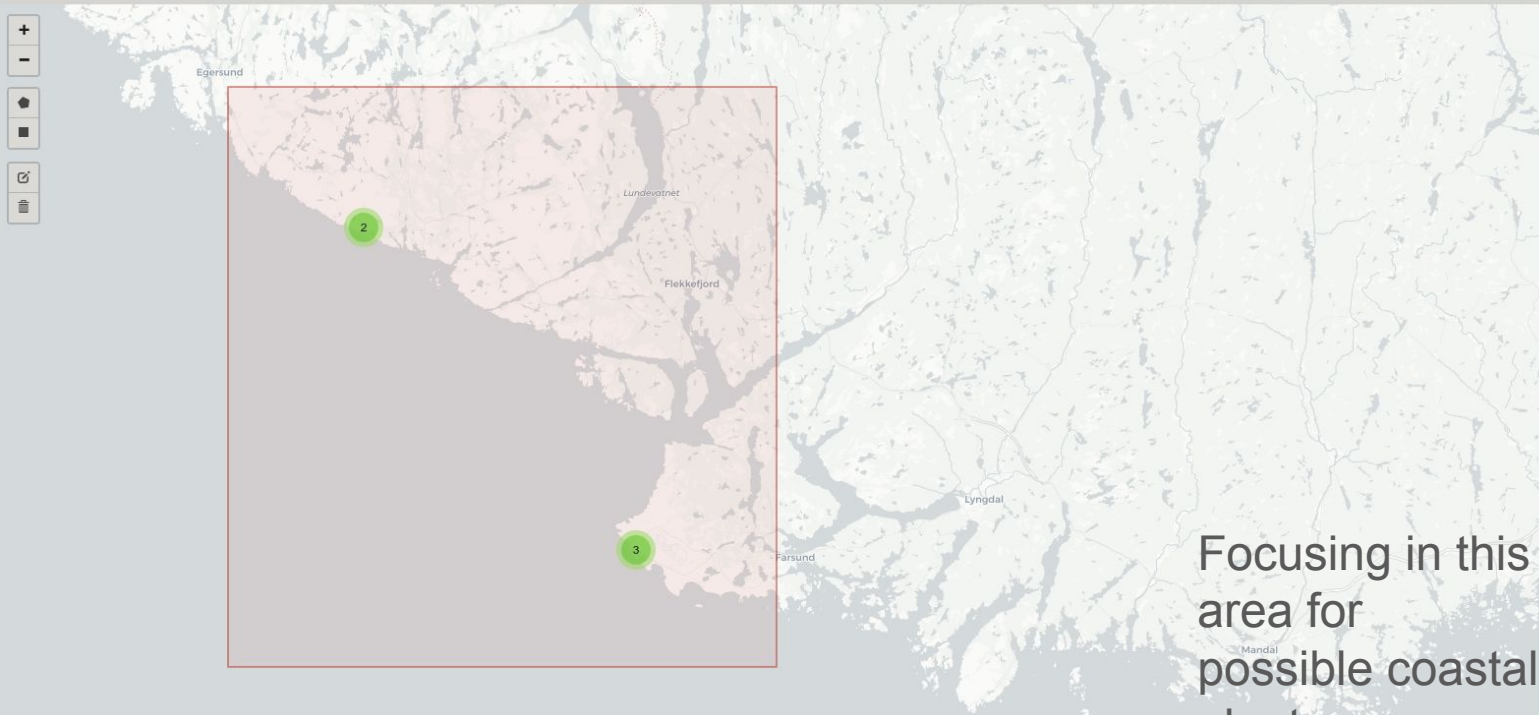
☐ Tax_division☐ Scientific_name☐ Tag1☐ Tag2

Home Overview Table Map

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 6.012, E 6.779, S 58.014, N 58.44 All points use original coordinates



Focusing in this
area for
possible coastal
plants

Sequence: OP265116.1

Ulva capillata voucher NO_085 ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit (rbcL) gene, partial cds; chloroplast.

Organism: [Ulva capillata](#)

Accession: OP265116

Mol Type: genomic DNA

Topology: linear

Base Count: 1313

Dataclass: STD

Tax Division: PLN

Location: 58.97 N 9.87 E

Specimen Voucher: NO_085

Country: Norway: Nevlunghavn

Show More



General

View: [EMBL](#) [FASTA](#)

Download: [EMBL](#) [FASTA](#)

Publications: [Show](#)

ORCID Data Claims: [Show](#)

Sequence Versions: [View](#)

ENA reveals
that these are
Ulva records for
the *rbcL* marker

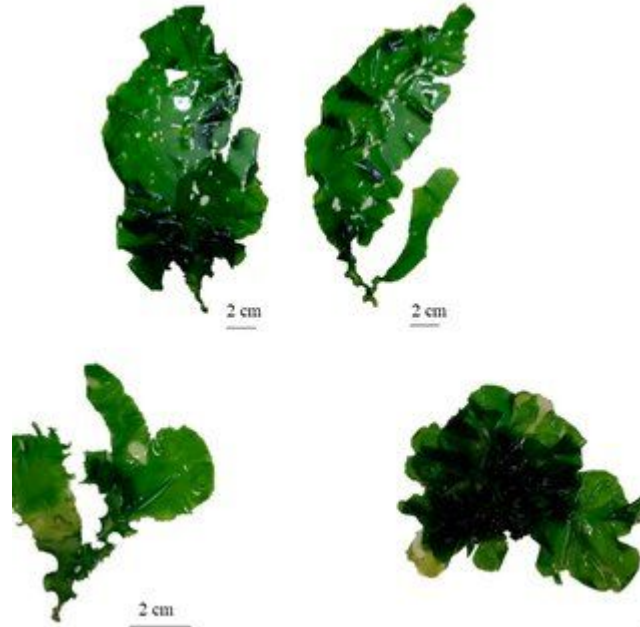
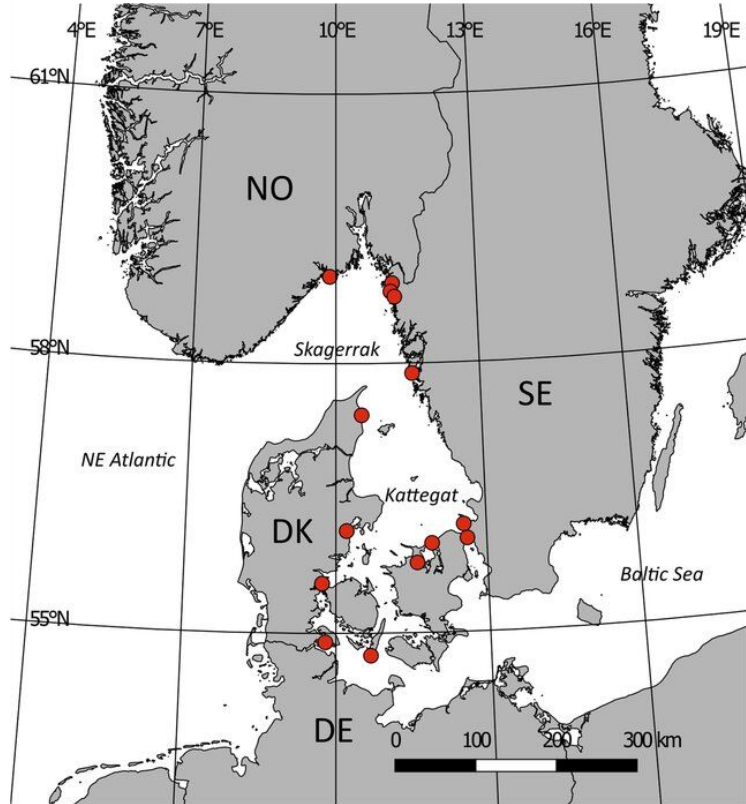


The European Nucleotide Archive (ENA) is part of the ELIXIR infrastructure
The ENA is an ELIXIR Core Data Resource. [Learn more](#)



The European Nucleotide Archive (ENA) is a Global Core Biodata Resource
The ENA is a GBC Global Core Biodata Resource. [Learn more](#)

Tubular greens: *Ulva* sp.



Steinhagen, S., Kramár, L. & Toth, G.B. The unheeded existence of the tubular greens: molecular analyses reveal the distribution of a new *Ulva* species (Ulvophyceae, Chlorophyta), *Ulva capillata* sp. nov. in the Atlantic-Baltic Sea transect. *J Appl Phycol* 35, 509–522 (2023). <https://doi.org/10.1007/s10811-022-02886-w>

[Home](#) [Overview](#) [Table](#) [Map](#)

Input data sources

☒ ENA☒ GBIF

Country of interest:

Norway

Dates of interest:

2023-05-04

to

2025-06-20

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE

Load Data

Table options

☐ Show filter

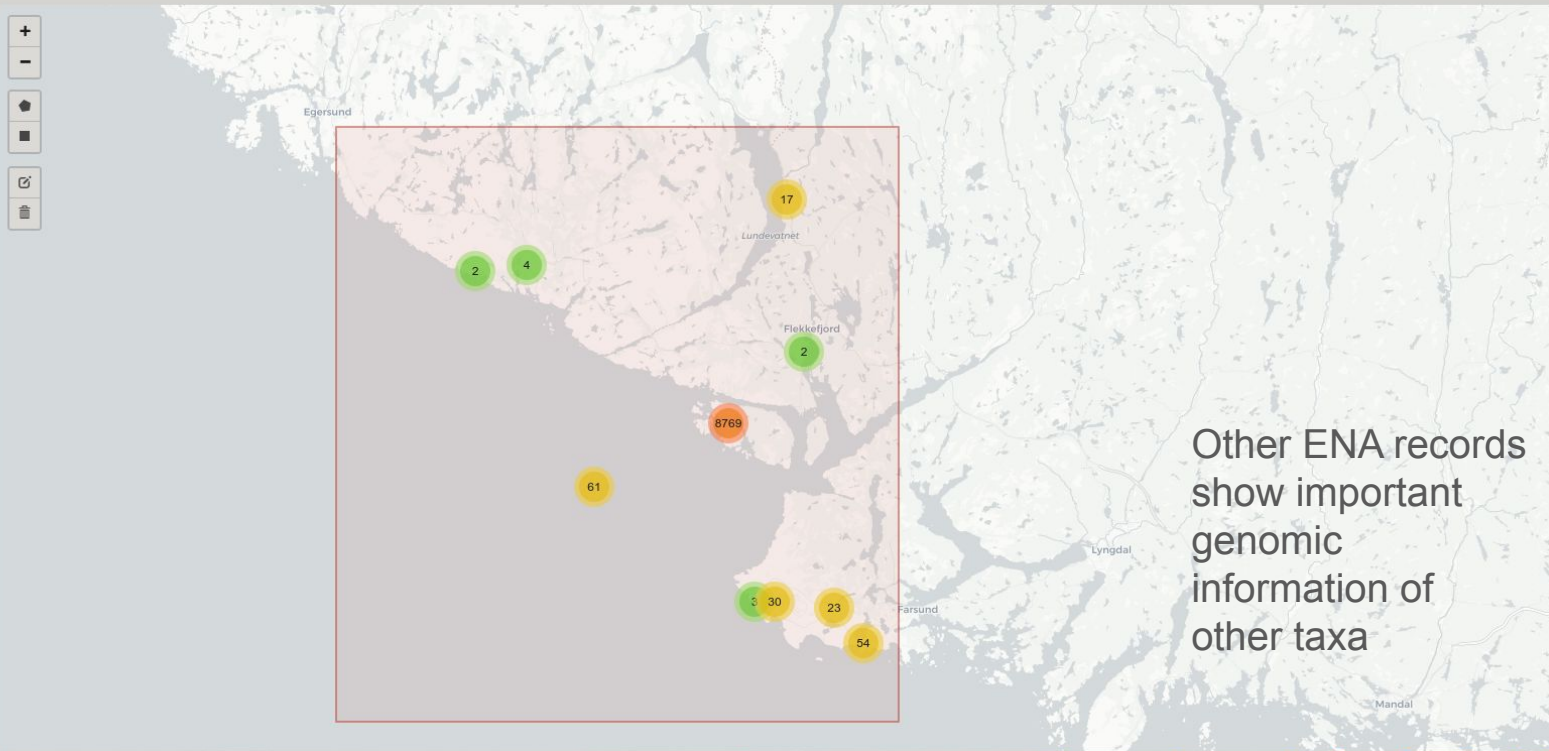
Group by

☐ Tax_division

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 6.012, E 6.779, S 58.014, N 58.44 All points use original coordinates



Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Norway

Dates of interest:

2023-05-04 to 2025-06-20

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE

Load Data

Table options

☐ Show filter

Group by

☐ Tax_division

ENA GBIF

accession	scientific_name	country	first_public	basis_of_record	decimalLatitude	decimalLongitude	tax_division2	phylum	class	order	family	genus	species
5172444168	Mycophaga testacea	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Anthomyiidae	Mycophaga	Mycophaga testacea
5172444420	Phaonia errans	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Muscidae	Phaonia	Phaonia errans
5172445003	Scaptomyza pallida	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Drosophilidae	Scaptomyza	Scaptomyza pallida
5172445008	Limnophyes natalensis	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Chironomidae	Limnophyes	Limnophyes natalensis
5172445778	Sylvicola cinctus	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Anisopodidae	Sylvicola	Sylvicola cinctus
5172448093	Cordyla crassicornis	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Mycetophilidae	Cordyla	Cordyla crassicornis
5172448853	Psychoda phalaenoides	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Psychodidae	Psychoda	Psychoda phalaenoides
5172449012	Bradysia irmleri	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Sciaridae	Bradysia	Bradysia irmleri
5172450816	Psychoda phalaenoides	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Psychodidae	Psychoda	Psychoda phalaenoides
5172451860	Syrphoctonus strigator	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Hymenoptera	Ichneumonidae	Syrphoctonus	Syrphoctonus strigator
5172451939	Zygomya	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Mycetophilidae	Zygomyia	
5172453849	Psychoda minuta	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Psychodidae	Psychoda	Psychoda minuta
5172453963	Trichocera	Norway	2024-10-01	MATERIAL_SAMPLE	58.228545	6.547146	Animalia	Arthropoda	Insecta	Diptera	Trichoceridae	Trichocera	Trichocera

Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Norway

Dates of interest:

2023-05-04 to 2025-06-20

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

MATERIAL_SAMPLE

Load Data

Table options

☐ Show filter

Group by

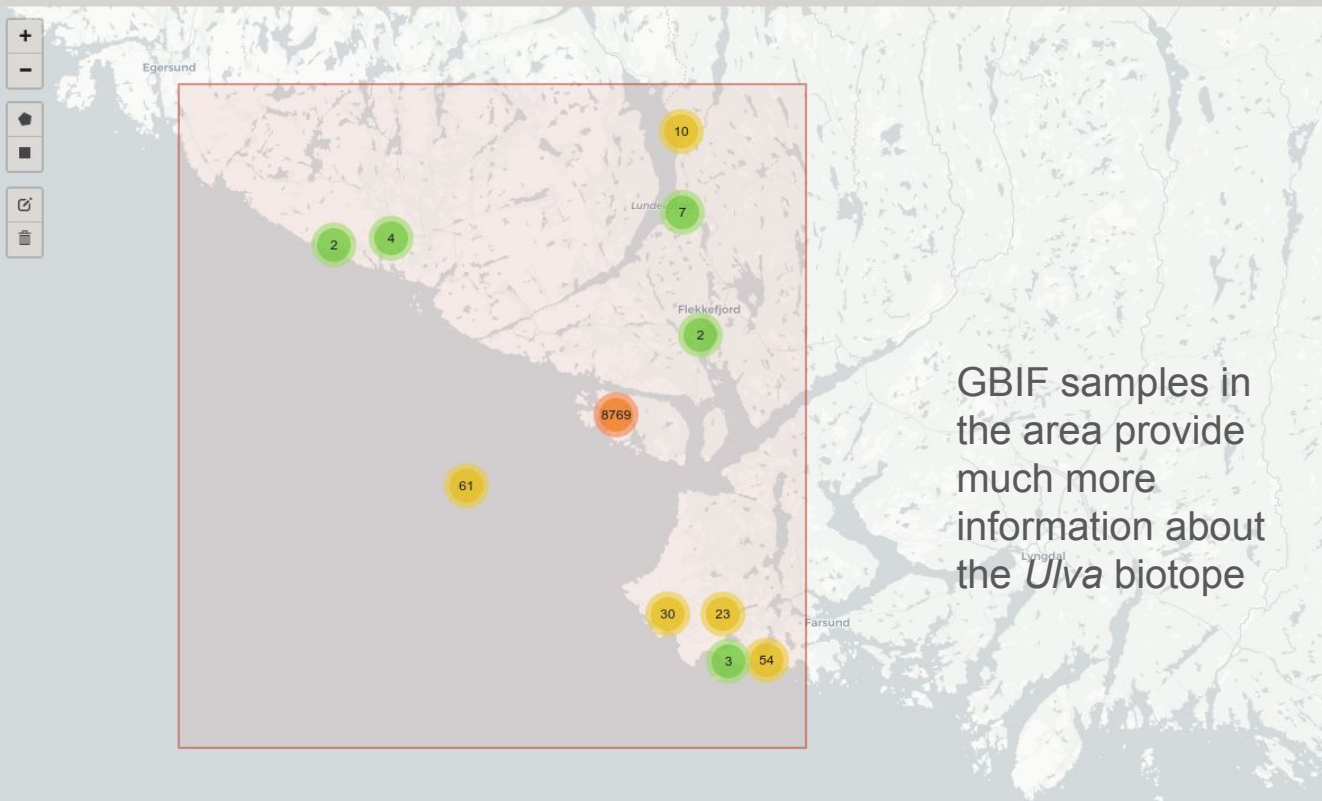
☐ Tax_division

Home Overview Table Map

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 6.012, E 6.779, S 58.014, N 58.44 All points use original coordinates



OCCURRENCE | 28 APRIL 2024 - 13 MAY 2024

Limnophyes natalensis (Kieffer, 1914)

Collected in Norway

Animalia > Arthropoda > Insecta > Diptera > Chironomidae > Limnophyes

DETAILS

Species: [Limnophyes natalensis \(Kieffer, 1914\)](#)

Location: Europe > Norway

Basis of record: Material sample

Event ID: 84415aa5-d19e-4c6f-af58-1e687a751d63

Dataset: [National insect monitoring in Norway](#)

Publisher: [Norwegian Institute for Nature Research](#)

Issues: [Footprint SRS invalid](#) [Country derived from coordinates](#)
[Continent derived from coordinates](#) [Taxon ID not found](#)



Record

Term

Interpreted

Original

Remarks

Basis of record

Material sample

MaterialSample

What we can do here:

- Compare all *rbcL* sequences for all *Ulva* records and other relative species and do phylogenetic studies and find spatial genetic patterns
 - Come up with possible evolutionary forces defining these patterns
 - Come up with relevant conservation strategies
- Compare spatial patterns of different taxa in the same region using ENA records and the relevant sequences
 - Identify persistent spatial patterns and elements that define them
 - Come up with relevant conservation strategies
- Evaluate specific habitats with the help of both ENA and GBIF records
 - Identify possible ecotypes of targeted taxa and describe possible adaptations
 - Evaluate environmental conditions in the habitats and identify threats
 - Conclude with conservation strategies to address these threats

The Norway *Ulva* use case

<!-- Demo as back-up slides ends here -->

The Cretan Sea use case

<!-- Demo as back-up slides starts here -->

127.0.0.1:6505

Exploring Molecular Biodiversity

Input data sources

☒ ENA
☐ GBIF

Country of interest:

Greece

Dates of interest:

2000-0 to 2026-0

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

Load Data

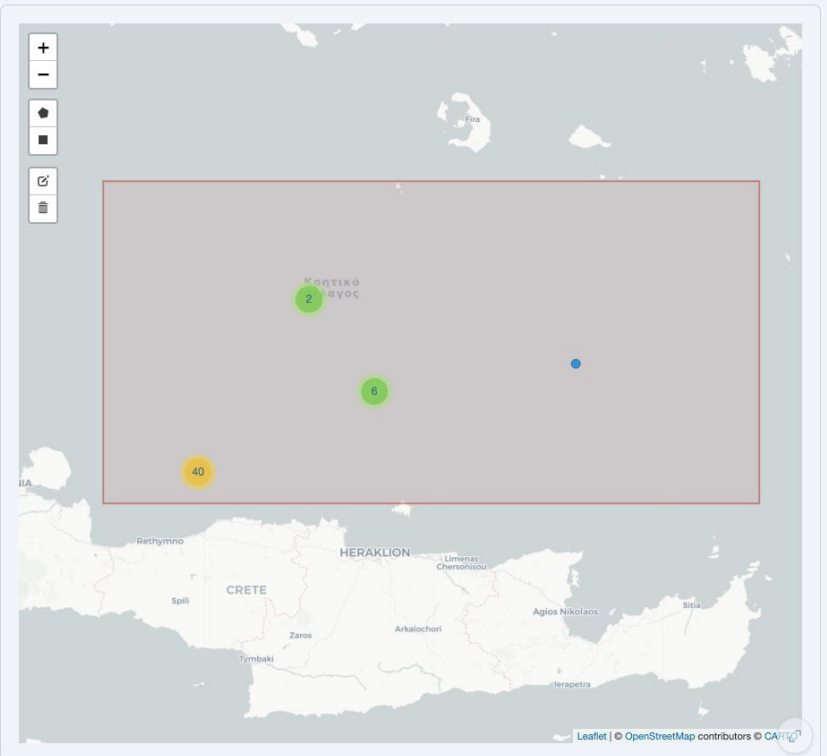
Table options

☐ Show filter

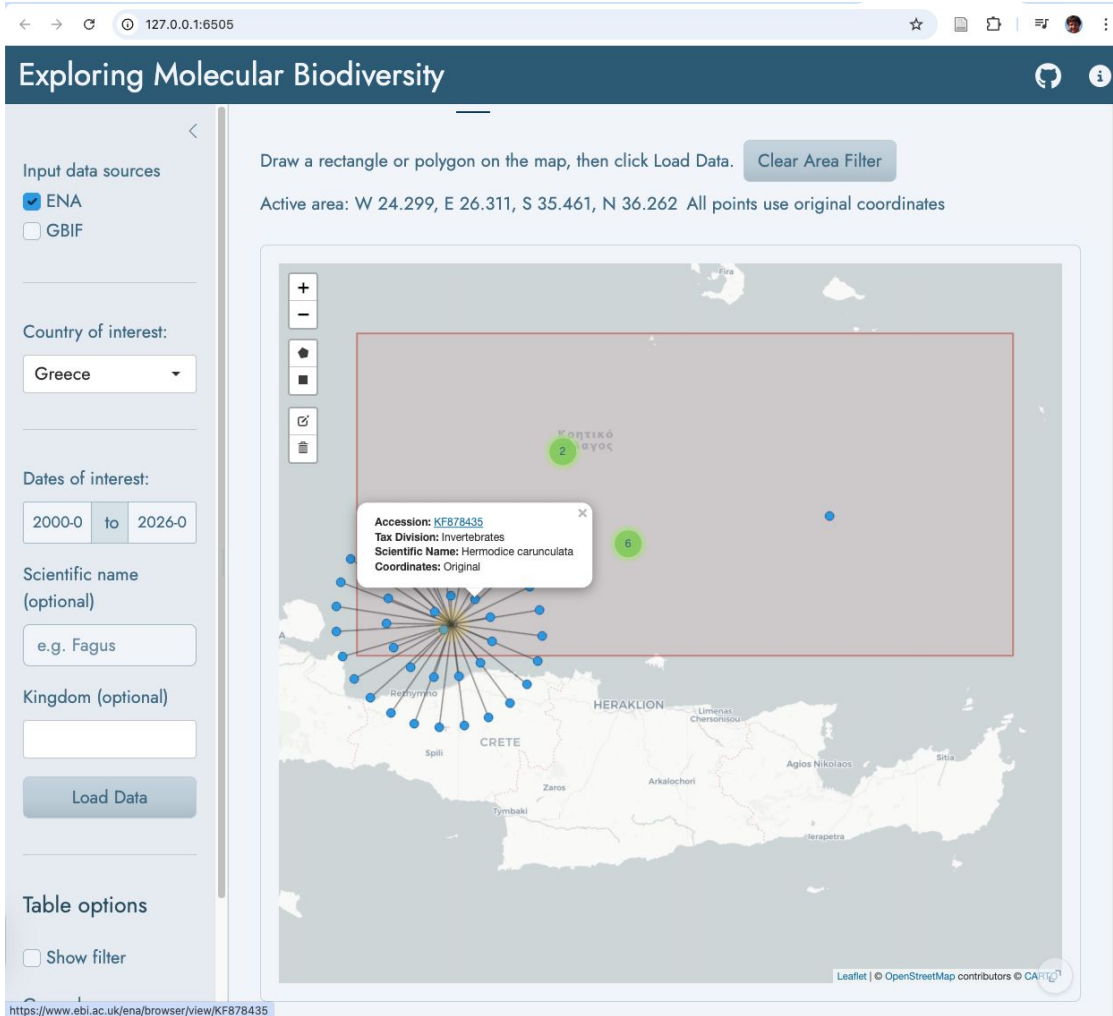
Group by

Draw a rectangle or polygon on the map, then click Load Data. Clear Area Filter

Active area: W 24.299, E 26.311, S 35.461, N 36.262 All points use original coordinates



Searching for ENA
sequence records between
2000-2026 in the **Cretan
Sea**



40 COI marker gene sequences of *Hermodice carunculata*

Hermodice carunculata, the **bearded fireworm**, is a type of **marine bristleworm** belonging to the **Amphinomidae** family, native to the tropical **Atlantic Ocean** and the **Mediterranean Sea**.



Source: https://en.wikipedia.org/wiki/Hermodice_carunculata, [https://en.wikipedia.org/wiki/File:Hermodice_carunculata_\(Bearded_Fireworm\).jpg](https://en.wikipedia.org/wiki/File:Hermodice_carunculata_(Bearded_Fireworm).jpg)

← → ↺ 🌐 ebi.ac.uk/ena/browser/view/KF878435 ☆ 📄 🗂 📧 👤 ⋮

🏠 EMBL-EBI home 🔍 Services 🧪 Research 👤 Training ⓘ About us 🔍 EMBL-EBI



European Nucleotide Archive

Home Submit Search About Support

Enter text search terms Search 🔍

Examples: Histone, BN000065

KF878435 View 🗄

Examples: Taxon:9606, BN000065, PRJEB402

Sequence: **KF878435.1** ⓘ

Hermodice carunculata voucher IMBBA NAGISA 1047 cytochrome c oxidase subunit I (COI) gene, partial cds; mitochondrial.

Organism: Hermodice carunculata (marine fireworm)

Accession: KF878435

Mol Type: genomic DNA

Topology: linear

Base Count: 607

Dataclass: STD

Tax Division: INV

Location: 35.54 N 24.59 E

Specimen Voucher: IMBBA NAGISA 1047

Country: Greece

Show More

General

📄 View: EMBL FASTA

📄 Download: EMBL FASTA

📄 Publications: Show

📄 ORCID Data Claims: Show

📄 Sequence Versions: View



The European Nucleotide Archive (ENA) is part of the ELIXIR infrastructure
The ENA is an ELIXIR Core Data Resource. [Learn more >](#)



The European Nucleotide Archive (ENA) is a Global Core Biodata Resource
The ENA is a GBC Global Core Biodata Resource. [Learn more >](#)

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I agree

ID KF878435; SV 1; linear; genomic DNA; STD; INV; 607 BP.
XX
AC KF878435;
XX
DT 16-DEC-2014 (Rel. 123, Created)
DT 16-DEC-2014 (Rel. 123, Last updated, Version 1)
XX
DE Hermodice carunculata voucher IMBBA NAGISA 1047 cytochrome c oxidase
DE subunit I (COI) gene, partial cds; mitochondrial.
XX
KW .
XX
OS Hermodice carunculata (marine fireworm)
OC Eukaryota; Metazoa; Spiralia; Lophotrochozoa; Annelida; Polychaeta;
OC Palpata; Aciculata; Eunicida; Amphinomidae; Hermodice.
OG Mitochondrion
XX
RN [1]
RP 1-607
RA Chatzigeorgiou G., Sarropoulou E., Vasileiadou K., Brown C., Faulwetter S.,
RA Kotoulas G., Arvanitidis C.;
RT "Population structure of Hermodice carunculata in Eastern Mediterranean Sea
RT (Crete)";
RL Unpublished.
XX
RN [2]
RP 1-607
RA Chatzigeorgiou G., Sarropoulou E., Vasileiadou K., Brown C., Faulwetter S.,
RA Kotoulas G., Arvanitidis C.;
RT ;
RL Submitted (20-NOV-2013) to the INSDC.
RL Biodiversity, Institute of Marine Biology, Biotechnology and Aquaculture,
RL Former American Base at Gournes, Heraklion, Crete 71003, Greece
XX
DR MD5; df67092da58424b8f7102910733c757b.

##Assembly-Data-START##
CC Sequencing Technology :: Sanger dideoxy sequencing
CC ##Assembly-Data-END##

Key Location/Qualifiers
FT source 1..607
FT /organism="Hermodice carunculata"
FT /organelle="mitochondrion"
FT /mol_type="genomic DNA"
FT /country="Greece"
FT /lat_lon="35.54 N 24.59 E"
FT /isolation_source="rocky shore"
FT /specimen_voucher="IMBBA NAGISA 1047"
FT /collected_by="Giorgos Chatzigeorgiou"
FT /collection_date="19-Sep-2007"
FT /dev_stage="mature"
FT /db_xref="taxon:759044"

KF87840	2014-12-16	Greece	-10	rocky shore	Hermodice carunculata	259844	35.54	24.59
KF87840	2014-12-16	Greece	-10	rocky shore	Hermodice carunculata	259844	35.54	24.59
KF87843	2014-12-16	Greece	-10	rocky shore	Hermodice carunculata	259844	35.54	24.59

127.0.0.1:6505

Exploring Molecular Biodiversity

Input data sources

- ☒ ENA
- ☐ GBIF

Country of interest:

Greece

Dates of interest:

2000-01- to 2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

Load Data

Table options

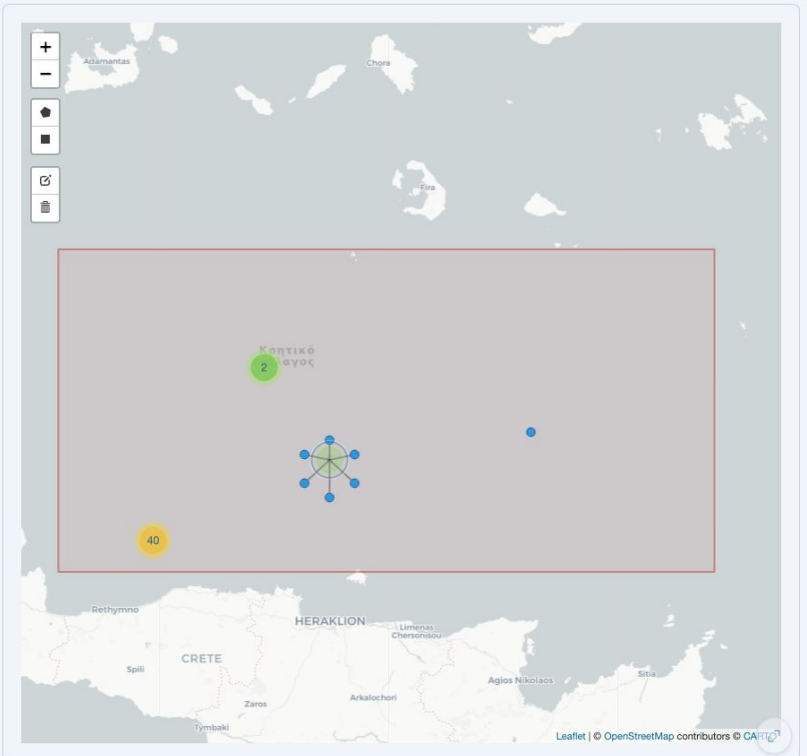
- ☐ Show filter

Group by

- ☐ Tax_division
- ☐ Scientific_name

Draw a rectangle or polygon on the map, then click Load Data. Clear Area Filter

Active area: W 24.299, E 26.311, S 35.461, N 36.262 All points use original coordinates



2 *Prionace glauca* (blue shark mitochondrial D-loop sequences)

6 16S rRNA gene sequences of copepods

1 ND1 sequence of *Pelophylax cretensis* (Cretan frog an endangered species, **endemic to the Greek island of Crete**)

127.0.0.1:6505

Exploring Molecular Biodiversity

Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Greece

Dates of interest:

2000-01- to 2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

HUMAN_OBSERVATION

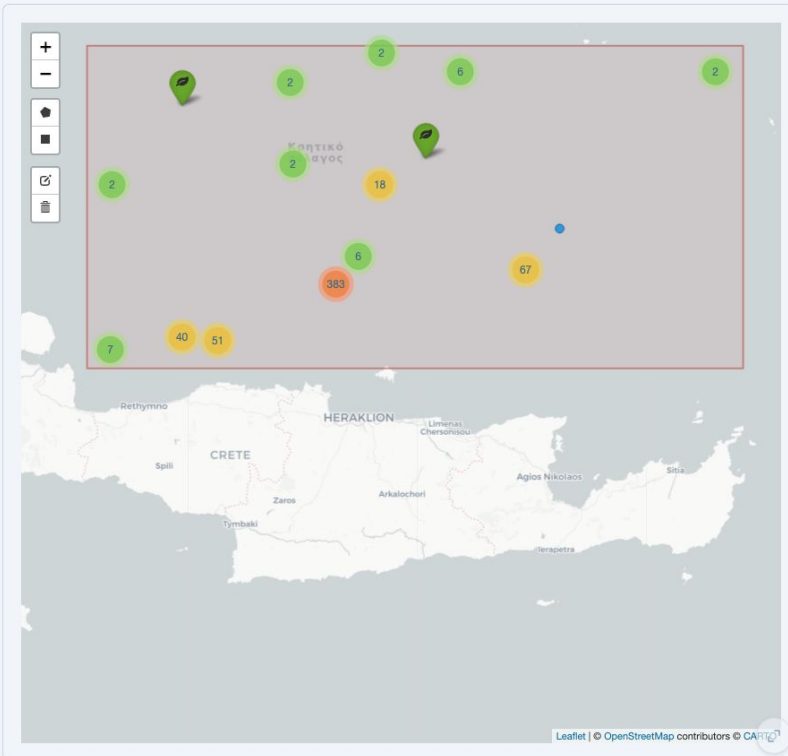
Load Data

Table options

☐ Show filter

Draw a rectangle or polygon on the map, then click Load Data. Clear Area Filter

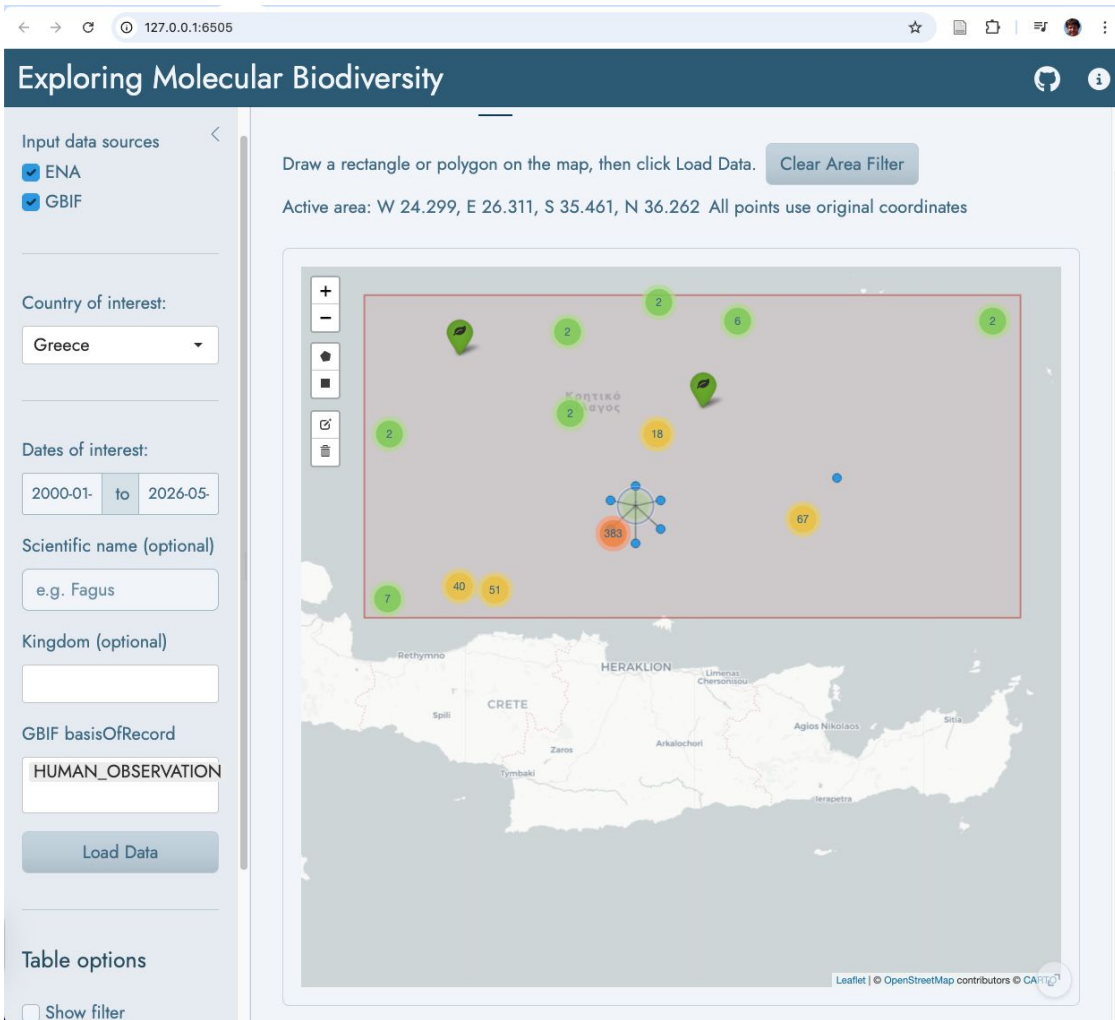
Active area: W 24.299, E 26.311, S 35.461, N 36.262 All points use original coordinates



Leaflet | © OpenStreetMap contributors © CARTO

What do GBIF human observations report for the same region?

<!-- like in the previous use cases, GBIF queries are restricted in terms of area due to the possible high number of results -->



A few hundreds of observations relatively close to the copepods

Exploring Molecular Biodiversity

Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Greece

Dates of interest:

2000-01- to 2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

HUMAN_OBSERVATION

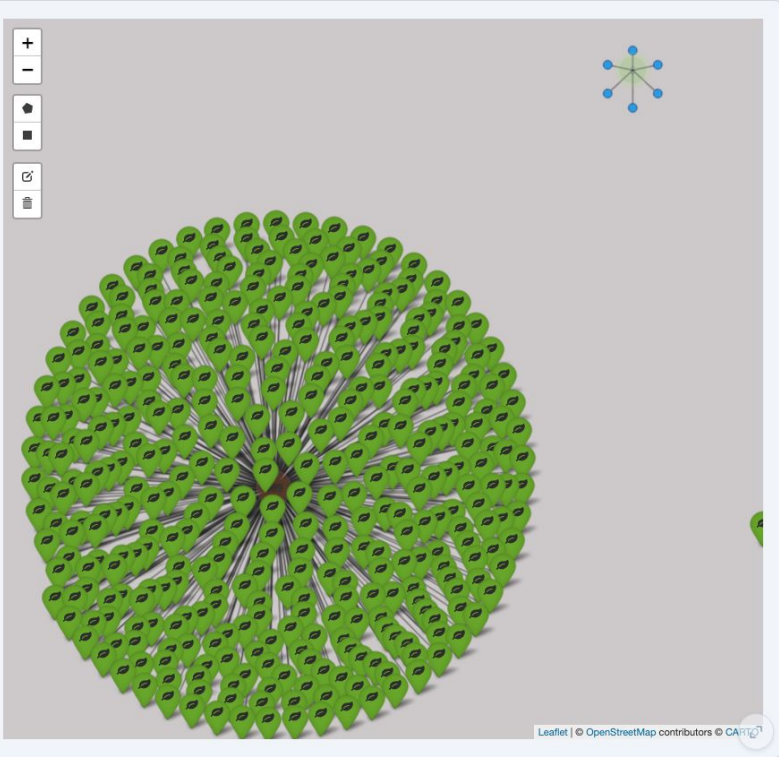
Load Data

Table options

☐ Show filter

Draw a rectangle or polygon on the map, then click Load Data. Clear Area Filter

Active area: W 24.299, E 26.311, S 35.461, N 36.262 All points use original coordinates



A few hundreds of observations relatively close to the copepods

inaturalist.org/observations/267504488

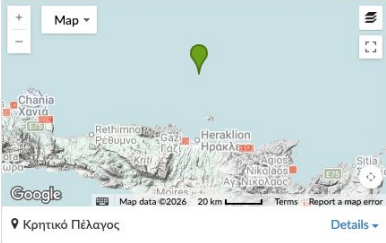
iNaturalist Explore Community More [Donate](#) Log In or Sign Up

Sweet Pea Shrub (*Polygala myrtifolia*) Research Grade



micha3006gu
444 observation

Observed: 1 Oct 2017 · 16:09 EEST
Submitted: 29 Mar 2025 · 15:44 EET



[Details](#)

Activity

micha3006gu suggested an ID ID Withdrawn 1y

 **Milkwort**
Genus: *Polygala*

d_pinto suggested an ID Improving 9mo

 **Sweet Pea Shrub**
Polygala myrtifolia

micha3006gu suggested an ID 9mo

 **Sweet Pea Shrub**

Community Taxon

[What's this?](#)

Sweet Pea Shrub (*Polygala myrtifolia*)
Cumulative IDs: 2 of 2

0 2/3rds 2

[Agree](#) [About](#)

Top Identifiers of Sweet Pea Shrub

Copyright Info and More

Observation © micha3006gu · [some rights reserved](#)

Plants included

Exploring Molecular Biodiversity



Input data sources

☒ ENA☒ GBIF

Country of interest:

Greece

Dates of interest:

2000-01-

to

2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

HUMAN_OBSERVATION

Load Data

Table options

☒ Show filter

Home Overview Table Map

Record filtering as an interactive mitigation

ENA

GBIF

accession	scientific_name	country	first_public	basis_of_record	decimalLatitude	decimalLongitude	tax_division2	phylum	class	order ↑	family	genus	species
							animalia			perciformes			
5935996639	Epinephelus marginatus	Greece	2025-08-17	HUMAN_OBSERVATION	35.590554	25.181104	Animalia	Chordata		Perciformes	Serranidae	Epinephelus	Epinephelus marginatus
5936315334	Epinephelus marginatus	Greece	2025-08-15	HUMAN_OBSERVATION	35.473959	25.085739	Animalia	Chordata		Perciformes	Serranidae	Epinephelus	Epinephelus marginatus
4440763302	Epinephelus marginatus	Greece	2023-10-27	HUMAN_OBSERVATION	35.563065	25.089114	Animalia	Chordata		Perciformes	Serranidae	Epinephelus	Epinephelus marginatus
5008383814	Epinephelus marginatus	Greece	2023-07-03	HUMAN_OBSERVATION	35.575719	24.761737	Animalia	Chordata		Perciformes	Serranidae	Epinephelus	Epinephelus marginatus
4015273674	Dentex dentex	Greece	2019-10-20	HUMAN_OBSERVATION	35.545172	25.149568	Animalia	Chordata		Perciformes	Sparidae	Dentex	Dentex dentex
4014867675	Dentex dentex	Greece	2019-10-12	HUMAN_OBSERVATION	35.460868	24.806427	Animalia	Chordata		Perciformes	Sparidae	Dentex	Dentex dentex

338 perciformes
observations



338

Exploring Molecular Biodiversity

Input data sources

- ☒ ENA
- ☒ GBIF

Country of interest:

Greece

Dates of interest:

2000-01- to 2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

HUMAN_OBSERVATION

Load Data

Table options

- ☒ Show filter

Home Overview Table Map

Similarly for other taxonomic groups of interest

ENA

GBIF

accession	scientific_name	country	first_public	basis_of_record	decimalLatitude	decimalLongitude	tax_division2	phylum	class	order ↑	family	genus	species
							animalia			cetacea			
3085365767	Delphinidae	Greece	2007-09-12	HUMAN_OBSERVATION	36.18602	26.28485	Animalia	Chordata	Mammalia	Cetacea	Delphinidae		
3085365653	Stenella coeruleoalba	Greece	2007-09-08	HUMAN_OBSERVATION	36.20095	25.32912	Animalia	Chordata	Mammalia	Cetacea	Delphinidae	Stenella	Stenella coeruleoalba
3085365792	Delphinidae	Greece	2007-06-08	HUMAN_OBSERVATION	36.21041	26.16845	Animalia	Chordata	Mammalia	Cetacea	Delphinidae		

1—3 of 3 rows Show 100 ▾

Previous 1 of 1 Next

Download as CSV

Exploring Molecular Biodiversity



Input data sources

☐ ENA☒ GBIF

Country of interest:

Greece

Dates of interest:

2000-01- to 2026-05-

Scientific name (optional)

e.g. Fagus

Kingdom (optional)

GBIF basisOfRecord

HUMAN_OBSERVATION

Load Data

Table options

☒ Show filter

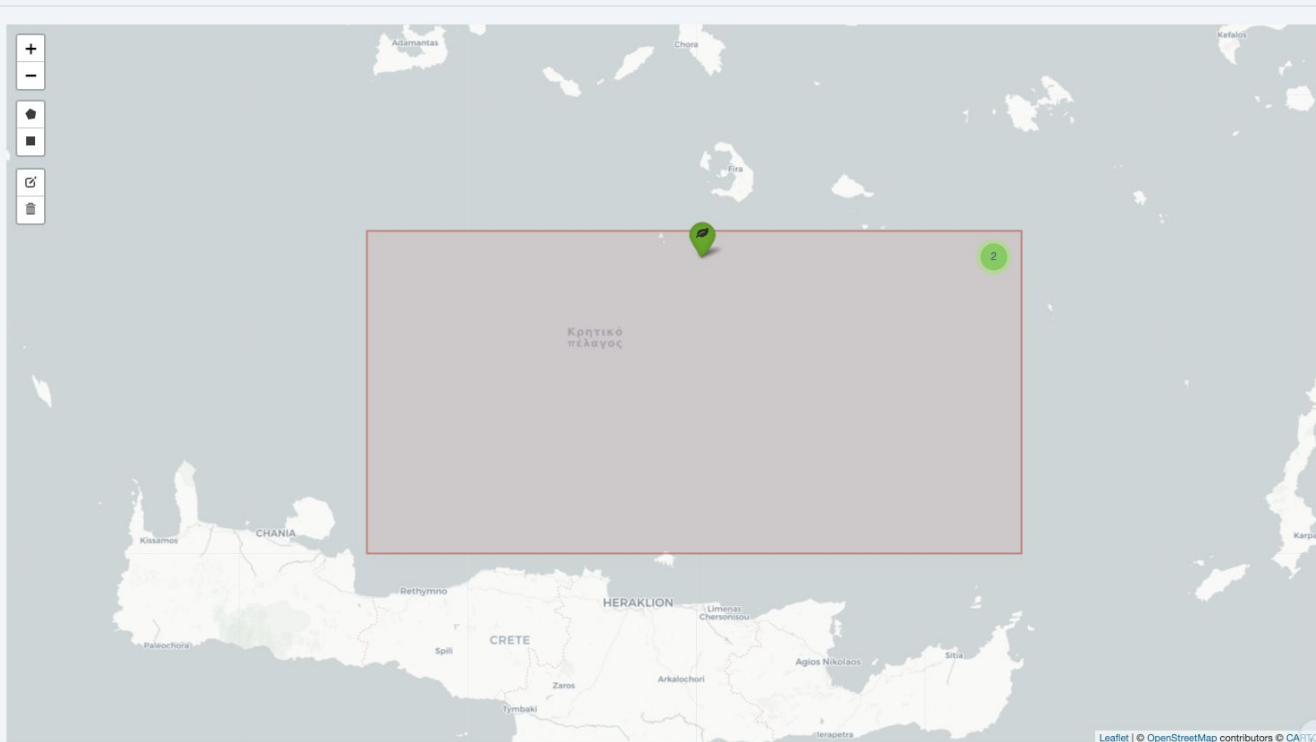
Home Overview Table Map

Cetacea observations

Draw a rectangle or polygon on the map, then click Load Data.

Clear Area Filter

Active area: W 24.299, E 26.311, S 35.461, N 36.262 All points use original coordinates



The Cretan Sea use case

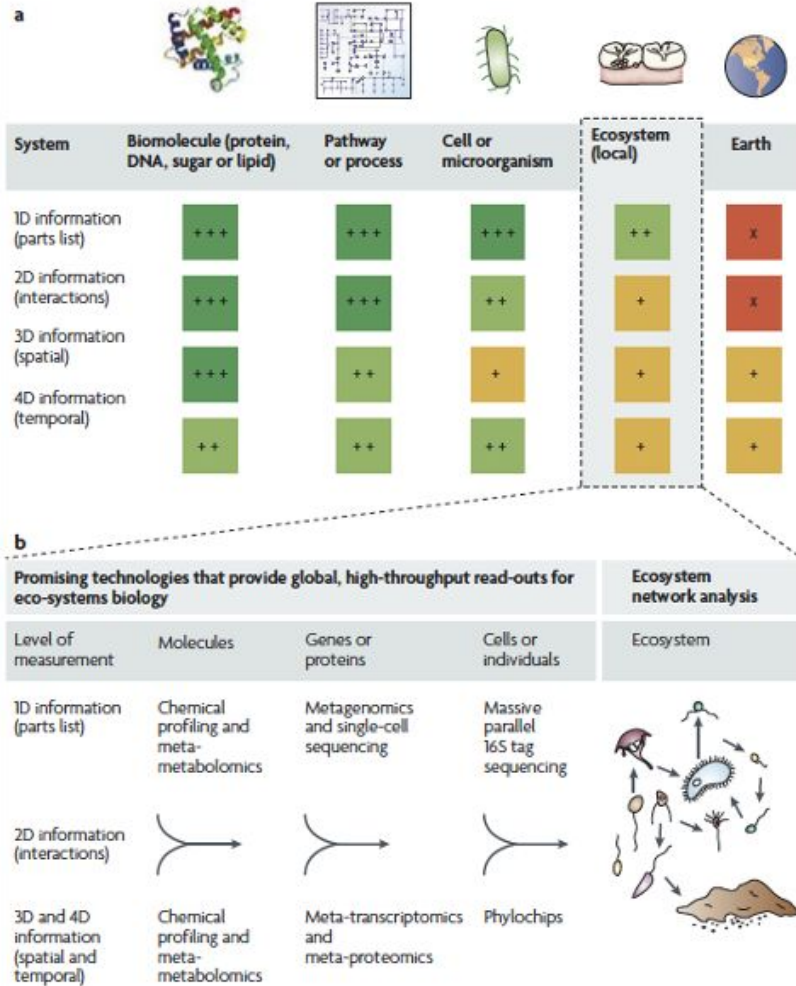
<!-- Demo as back-up slides ends here -->

Remarks

- Explore what is available for a time period/geographic area
 - ENA sequences
 - GBIF deposited observations (a number of types)
- Compare patterns
 - Identify more than ones types of evidence for the presence of a taxonomic group (not shown in this demo)
 - Spot incorrect annotations, confirm correct ones
 - Explore co-presence
- Interactive querying, filtering, and plotting
 - User interface features assist in the navigation of biodiversity data (molecular and beyond)
- Motivation for high quality metadata annotations

Systems Biology: from proteins to environments

Source: **Molecular eco-systems biology: towards an understanding of community function.** Raes J. and Bork P. (2008) Nat. Rev. Microbiol., 2008, 6, 693–699.



Inspiration: **Eco-systems biology**

Odyssey

1D: visualizing the parts

3D and 4D: geographical (map-based) and timespan query

+, a few scattered non-high-throughput data sets available and model building is restricted to case studies;

x, almost no data available).

BFSP WP 2027 - 2028

2D: (biotic) interactions
FAIR Data, Task 3, lead
Dr. Patrick Ruch)

Food for thought, heading to the next agenda item

- Features to add/improve
 - Import/show custom user datasets
 - Import layers
 - Earth surface/bathymetry satellite data
 - Natura 2000 - Marine Protected Areas
 - Land usage
 - Environmental feature data maps
 - Export data/layers/sequences retrieval
 - LLMs to better translate location names into (approximate) coordinates, or to flag records (e.g. according to the sequence type, like marker gene, plastome, ...)
- Involve more countries and ELIXIR nodes to cover larger areas

Acknowledgements

- ELIXIR BFSP co-chairs & hub personnel
Physilia, Katharina, Tereza, Cyril, Rob
- Non-Odyssey-team colleagues @CERTH,
@DUTH, GR; @NMBU, NO
- Dag Endresen, GBIF Norway
- To the API providers (ENA, GBIF) without
whose resources, and documentation, an
approach like Odyssey would not have
been possible



Grefsenkollveien 100, 0490 Oslo, Norway

59°57'32.0"N 10°48'12.5"E (59.95889, 10.80346)

March, 4th, 2026, 18:19

Questions / Feedback / Discussion

Feedback on slido: #138 60 53

<https://app.sli.do/event/1wEhd8h5mR1w7XFVnx6F18>





Thank you



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[/company/elixir-europe](https://www.linkedin.com/company/elixir-europe)

elixir-europe.org

Group photograph



BFSP Open Call

Odyssey

Connecting molecular and geographical biodiversity data

Hands-on training

May 18th 2026



CERTH
CENTRE FOR
RESEARCH & TECHNOLOGY
HELLAS



UiO : University of Oslo



ΔΗΜΟΚΡΙΤΕΙΟ
ΠΑΝΕΠΙΣΤΗΜΙΟ
ΘΡΑΚΗΣ
DEMOCRITUS
UNIVERSITY
OF THRACE



Discussion points

- [NStrepis] add checkbox next to each item in the table, make it possible to select which to a. view on the map, and b. **download**
- [NStrepis] How to integrate/cross-link Odyssey to ELIXIR actions/resources?
- [NStrepis] Can LLMs improve the filtering capabilities, is there an agentic way to follow?
- [NStrepis] Is there a plan on how to promote Odyssey and increase its adoption?

R for macOS

This directory contains binaries for the base distribution and of R and packages to run on macOS. R and package binaries for R versions older than 4.0.0 are only available from the [CRAN archive](https://cran.r-project.org) so users of such versions should adjust the CRAN mirror setting (<https://cran.r-project.org>) accordingly.

Note: Although we take precautions when assembling binaries, please use the normal precautions with downloaded executables.

R 4.6.0 "Because it was There" released on 2026/04/24

Please check the integrity of the downloaded package by checking the signature:

```
pkgutil --check-signature R-4.6.0-arm64.pkg
```

in the *Terminal* application. If Apple tools are not available you can check the SHA1 checksum of the downloaded image:

```
openssl sha1 R-4.6.0-arm64.pkg
```

Latest release:

For Apple silicon (M1,2,...) Macs:

[R-4.6.0-arm64.pkg](#)

SHA1-

hash: 85775cbe3ba6242d2a2f3693a76422e3303bda9c
(ca. 101MB, notarized and signed)

R 4.6.0 binary for macOS 14 (**Sonoma**) and higher for arm64 Macs, signed and notarized packages.

Contains R 4.6.0 framework, R.app GUI 1.83, Tcl/Tk 8.6.12 X11 libraries and Texinfo 7.3. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `tcltk` R package or build package documentation from sources.

If installing from the *Downloads* folder you must allow the Apple Installer access to that directory. Denying the access will result in you not being able to install any software from there and then you must move the file to a location that is not private such as your home folder.

Note: the use of X11 (including `tcltk`) requires [XQuartz](#) (version 2.8.5 or later). Always re-install XQuartz when upgrading your macOS to a new major version.

This release uses Xcode 26.0 (the same LLVM toolchain up to 26.3) and GNU Fortran 14.2. If you wish to compile R packages which contain Fortran code, you may need to download the corresponding GNU Fortran compiler from <https://mac.R-project.org/tools>. Any external libraries and tools are expected to live in `/opt/R/arm64`.

For older Intel Macs:

[R-4.6.0-x86_64.pkg](#)

SHA1-

hash: e6430cd1b7b27e83597edcd95d0a828228b701a
(ca. 104MB, notarized and signed)

R 4.6.0 binary for macOS 11 (**Big Sur**) and higher, signed and notarized packages.

Contains R 4.6.0 framework, R.app GUI 1.83, Tcl/Tk 8.6.12 X11 libraries and Texinfo 7.3. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `tcltk` R package or build package documentation from sources.

If installing from the *Downloads* folder you must allow the Apple Installer access to that directory. Denying the access will result in you not being able to install any software from there and then you must move the file to a location that is not private such as your home folder.

Quick note for Mac OS X users

- Update to R v4.6.0 recommended
<https://cran.r-project.org/bin/macosx/>
- XQuartz might have to be updated too
xquartz.macosforge.org Administrator access required)



The XQuartz project is an open-source effort to develop a version of the [X.Org X Window System](#) that runs on macOS. Together with supporting libraries and applications, it forms the X11.app that Apple shipped with OS X versions 10.5 through 10.7.

Quick Download

Download	Version	Released	Info
XQuartz-2.8.5.pkg	2.8.5	2023-01-26	For macOS 10.9 or later

License Info

An XQuartz installation consists of many individual pieces of software which have various licenses. The X.Org software components' licenses are discussed on the [X.Org Foundation Licenses page](#). The `quartz-wm` window manager included with the XQuartz distribution uses the [Apple Public Source License Version 2](#).